

Global Memory Tech

Weekly theme: 3Q DRAM ASP rising well, memory upside with TSMC/ASML guidance

Industry Overview

What we learned from our 3Q contract price check

Investors have increasingly asked about memory chipmakers' likely 3Q earnings miss, given limited ASP hike vs consensus. This prompted us to check newly negotiated 3Q DRAM contract prices more closely. What we learned: (1) stronger server DRAM pricing, up 20-30% QoQ, led by high-speed LPDDR5 vs consensus of 20% or less; (2) upbeat spot demand, led by commodity DDR5 and even legacy DDR4, with July prices up again MoM; (3) a near-term rise in orders for higher-priced HBM4 vs cheaper HBM3e; (4) LTA-based DRAM sales well below 50% of total, with non-LTA portion at 60-70%, and a solid QoQ price rise (e.g., 5-10%), often settled even under LTAs; and (5) some OEMs' rush orders confirming more than 20% QoQ price rise, not only for commodity DRAM but also for NAND. Overall, our global forecast for 3Q DRAM ASP, up 21% QoQ, is consistent with what we learned from channel check. This is more optimistic than TrendForce's current assumptions of a 13-18% QoQ rise in 3Q conventional DRAM, or only 8-13% including HBM. Of course, TrendForce's 2Q DRAM ASP assumptions, up 58-63% QoQ for conventional DRAM only or 53-58% on an overall blended basis, including HBM, seem to be 10ppt higher than Korea-specific average but in line with ex-Korea global assumption.

Spot prices also rise, led by DRAM

DRAM spot prices have risen for eight consecutive weeks, though being at a abnormally high level. This is completely different vs 2-3 months ago sentiment (e.g., no further upside at US\$35-40 for 16Gb DDR5; already higher than HBM price). Many OEMs, which had previously purchased DRAM at only around US\$5 per unit, resisted the DRAM spot-price rally and consequently reduced PC/tablet/smartphone production volume. But, they are now buying more memory chips to prepare for higher sales in Sept. and the 4Q peak season. Since spot trading volume is very small, at a low-single-digit% of global supply, some OEMs are purchasing conventional DRAM on more of a rush-order basis. Current spot prices for both DDR5 and DDR4 point to a roughly 20% QoQ rise. NAND spot prices also recovered well this week, led by 1Tb, up 4% WoW. This is also a positive signal to expect 10%+ NAND ASP rise in 3Q. But, similar to Korea's 2Q DRAM ASP miss vs global trend, Japan NAND ASP should not have been upbeat vs Street expectation; still, it is too early to expect a 3Q miss, given solid July data, with monthly spot/contract prices rising.

Positive implications of TSMC's strong results/capex

TSMC's (Taiwan Semi Manufacturing Co.) strong 2Q results, with sales of US\$40bn (+34% YoY), and OPM of 60%; bullish guidance, including a 2nm node ramp-up and low-40% annual revenue growth; and record-high capex of US\$60-64bn for 2026, plus a separate additional US\$100bn investment in US fabs, have positive implications for memory, as TSMC's customers need more advanced HBM, SOCAMM, and eSSD. ASML's 2Q results/guidance were also favorable for memory due to muted China exports (local chipmakers' lower spending); more Taiwan/logic-centric sales growth (vs memory/Korea); and management's optimistic view on new AI chip and high NA EUV (memory order upside for new logic).

>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.

Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.

BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Refer to important disclosures on page 15 to 16.

12995153

Timestamp: 16 July 2026 10:09AM EDT

16 July 2026

Equity
Global
Technology

Simon Woo, CFA >>
Research Analyst
Merrill Lynch (Seoul)
+82 2 3707 0554
simon.woo@bofa.com

Dai Shen >>
Research Analyst
Merrill Lynch (Hong Kong)
dai.shen@bofa.com

Vivek Arya
Research Analyst
BofAS
vivek.arya@bofa.com

Mikio Hirakawa >>
Research Analyst
BofAS Japan
mikio.hirakawa@bofa.com

Matt Shin >>
Research Analyst
Merrill Lynch (Seoul)
matt.shin2@bofa.com

Exhibit 1: DDR5/DDR4 well rebounded this week; even 1Tb NAND recovery seen

Spot-market prices among DRAM and NAND

US\$	Current	WoW	QoQ	YoY
DRAM spot				
16Gb DDR5	49.2	3%	28%	712%
16Gb DDR4	80.1	3%	16%	829%
8Gb DDR4	39.9	6%	21%	685%
4Gb DDR4	12.9	4%	81%	426%
NAND spot				
1Tb wafer	25.1	4%	-3%	397%
512Gb wafer	19.2	-1%	-9%	617%
256Gb wafer	10.2	0%	-3%	578%

Source: DRAMeXchange

BofA GLOBAL RESEARCH

AI: Artificial intelligence

ASP: Average selling price

DDR4/5: 4th/5th gen double-data rate DRAM

DRAM: Dynamic random-access memory

EUUV: Extreme ultraviolet lithography

eSSD: Enterprise solid-state drive

Gb/Tb: Gigabit/Terabit

GM/OPM: Gross/OP margin

HBM: High bandwidth memory

HBM3e/4: 5th/6th gen of HBM

LPDDR5: Low-power DDR5

LTA: Long-term agreement

NAND: Not-AND memory

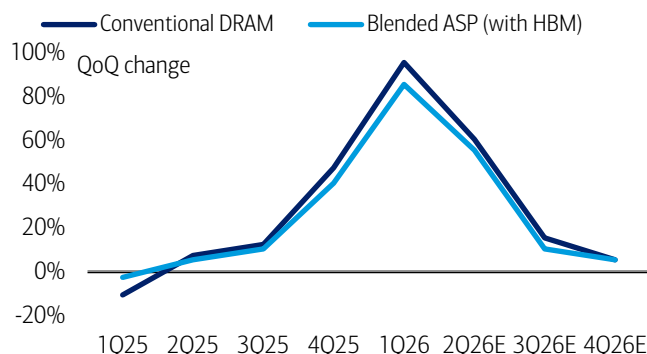
OEM: Original equipment manufacturer

SOCAMM: Small-outline compression attached memory module

TrendForce memory ASP forecast vs BofA estimates

Exhibit 2: TrendForce expects conventional DRAM price to moderate through 2H26, decelerating from ~93-98% QoQ growth in 1Q to ~58-63% in 2Q to ~13-18% in 3Q and just ~3-8% by 4Q

TrendForce DRAM ASP forecast updated as of 30th June 2026



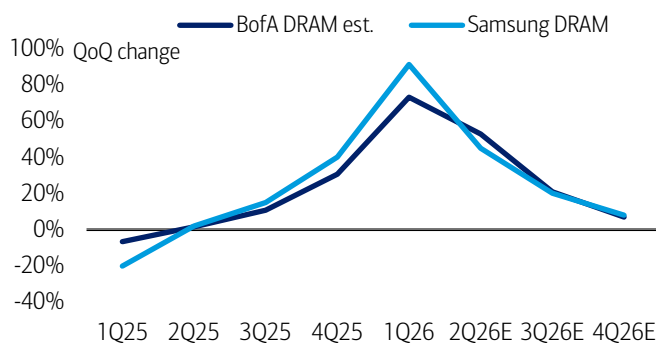
*Blended ASP refers to entire DRAM (conventional + HBM)

Source: TrendForce, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 4: Following the exceptional +70-90% QoQ increase in 1Q26, we expect global DRAM and Samsung ASP growth to moderate to +45-50% in 2Q and ~+20% QoQ in 3Q26 and high-single-digit QoQ in 4Q26

BofA Global DRAM ASP and Samsung DRAM ASP estimates

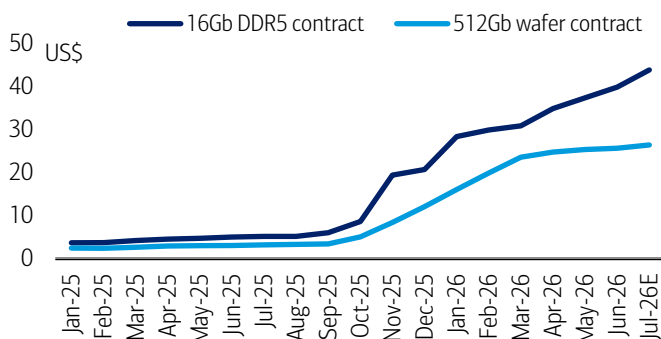


Source: Companies, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 6: Both DRAM and NAND prices are at record highs, with DDR5 DRAM reaching approximately \$45 compared with NAND at ~\$25

16Gb DDR5 vs 512Gb wafer contract price trend

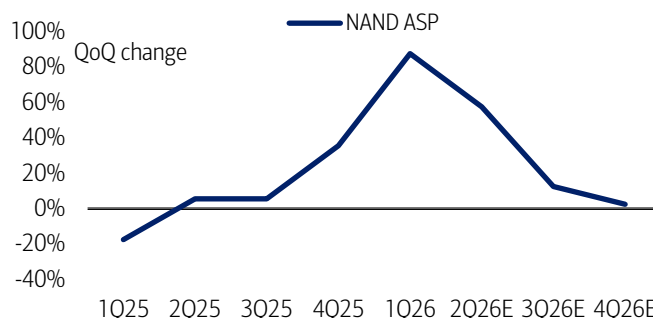


Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 3: TrendForce forecasts NAND ASP to normalize to ~+10-15% QoQ in 3Q26 and ~+0-5% QoQ in 4Q26 after reaching peak levels of ~85-90% QoQ and ~55-60% QoQ in 1Q/2Q26

TrendForce NAND ASP forecast updated as of 30th June 2026

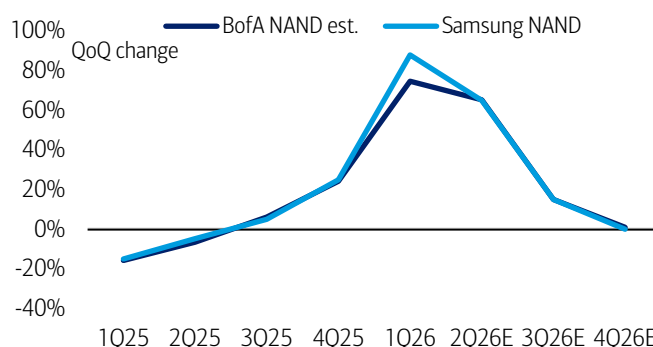


Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 5: Following the +75-88% QoQ spike in 1Q26, we expect global NAND and Samsung ASP growth to moderate to ~+65% QoQ in 2Q26, easing further to ~+15% QoQ in 3Q26 and near-flat levels by 4Q26

BofA Global NAND ASP and Samsung NAND ASP estimates

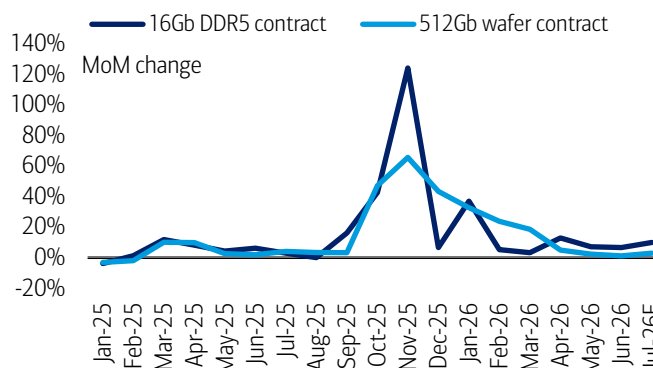


Source: Companies, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 7: NAND and DRAM price expected to rise 3-10% MoM in Jul-26

16Gb DDR5 vs 512Gb wafer contract price trend – MoM change



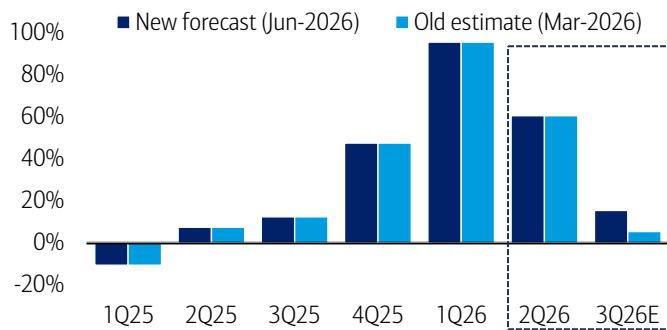
Source: TrendForce

BofA GLOBAL RESEARCH



Exhibit 8: TrendForce has revised up its 3Q26 DRAM ASP forecast to 13–18% QoQ vs previous estimate of 3–8% QoQ

TrendForce DRAM ASP forecast (QoQ change) – Jun-26 update vs Mar-26



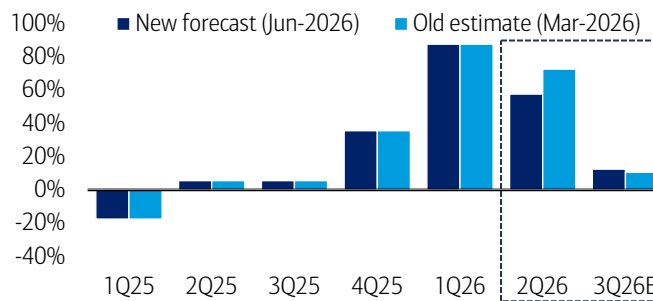
TrendForce's latest update as of 30th June-2026

Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 9: TrendForce has revised down its 2Q26 NAND ASP forecast to 55–60% QoQ (from 70–75%), while raising its 4Q26 outlook to 10–15% QoQ from the earlier 8–13% range

TrendForce NAND ASP forecast (QoQ change) – Jun-26 update vs Mar-26



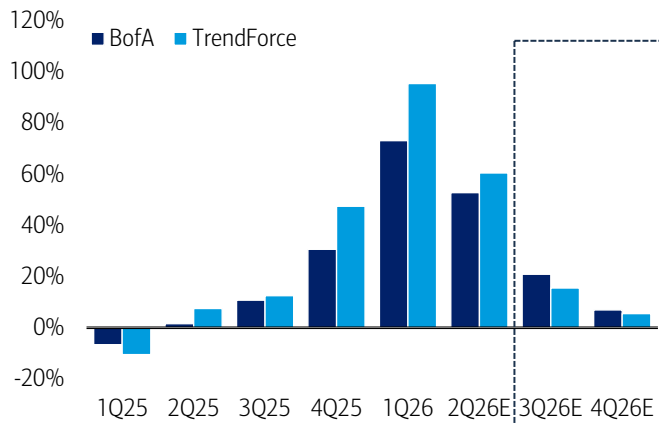
TrendForce's latest update as of 30th June-2026

Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 10: Our estimates are largely in line for 3Q26 and 4Q26 and more conservative for 2Q26 relative to TrendForce, with DRAM ASPs projected to increase 53% / 21% / 7% QoQ across 2Q–4Q26 (BofA)

BofA vs TrendForce DRAM ASP assumptions (QoQ change)



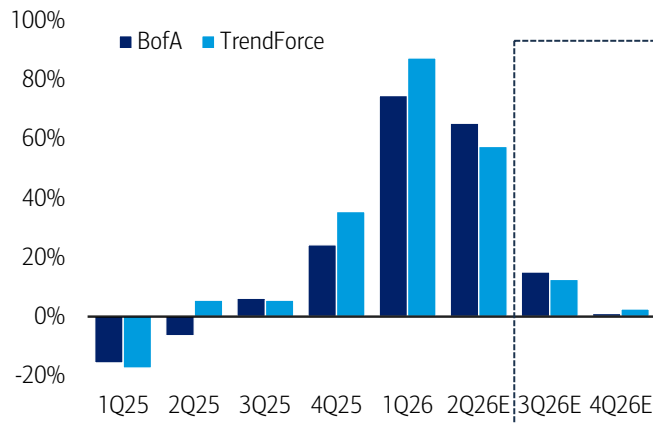
TrendForce's latest update as of 30th June-2026 vs BofA update as of 10th July 2026

Source: TrendForce, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 11: Similarly, our NAND assumptions are either conservative or broadly aligned with TrendForce, with ASPs expected to grow 65% / 15% / 1% QoQ across 2Q–4Q26

BofA vs TrendForce NAND ASP assumptions (QoQ change)



TrendForce's latest update as of 30th June-2026 vs BofA update as of 10th July 2026

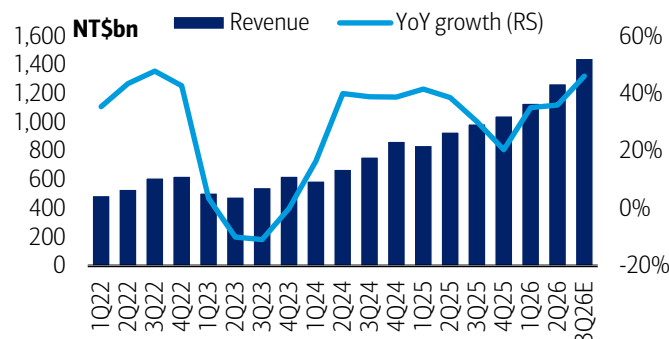
Source: TrendForce, BofA Global Research estimates

BofA GLOBAL RESEARCH

TSMC's strong 2Q results and higher capex guidance

Exhibit 12: 2Q26 sales (NT\$1,270bn; +36% YoY) came in at the high-end of its guidance; 3Q sales guidance (NT\$1,446bn) also indicates solid YoY growth (up 46%)

TSMC – Revenue and growth trend

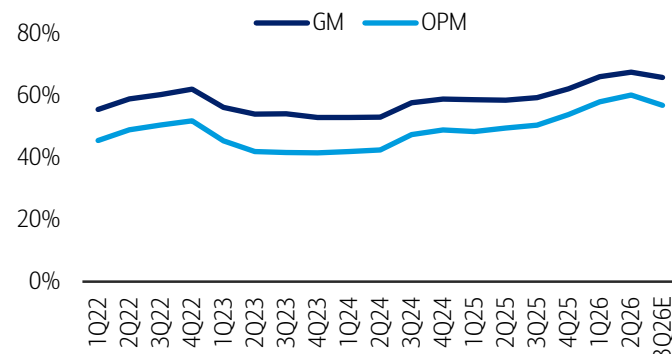


Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 13: Margins further improved in 2Q (GM 68%, OPM 60%); high-margin profile well-expected to continue

TSMC – Gross margin and OP margin trend

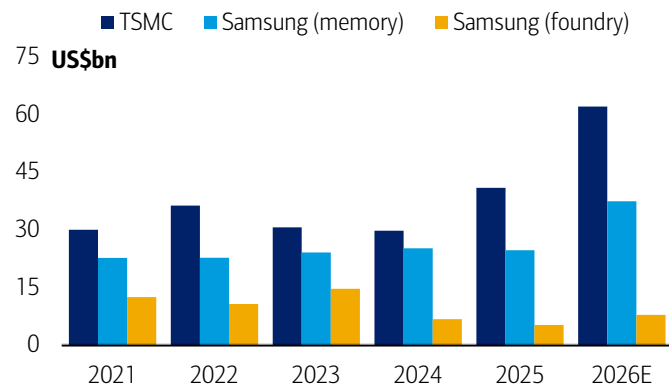


Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 14: TSMC revised up its 2026 annual capex guidance from US\$60-64bn vs US\$56bn previously

Capex comparison between TSMC and Samsung memory/foundry

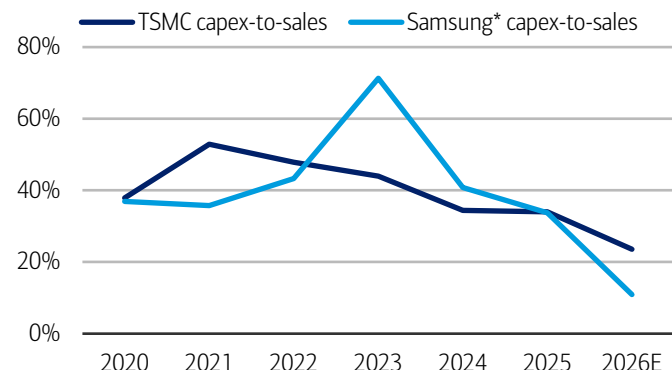


Source: Companies, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 15: Significant drop in capex-to-sales ratio seen for both TSMC and Samsung due to AI-driven exceptionally robust topline growth

Capex-to-sales comparison – TSMC and Samsung *memory-only



Source: Companies, BofA Global Research estimates

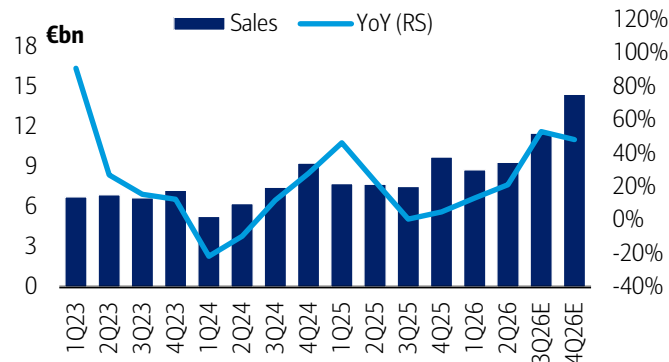
BofA GLOBAL RESEARCH



ASML's upbeat 2Q results and 2H guidance

Exhibit 16: Upbeat 2Q26 sales (€9.3bn; up 21% YoY) led by stronger-than-expected IBM (installed base management) sales; management's sales guidance indicates robust growth in 2H (3Q: €11.5bn, up 53% YoY / 4Q: €14.4bn, up 48% YoY)

ASML – Total sales and growth trend

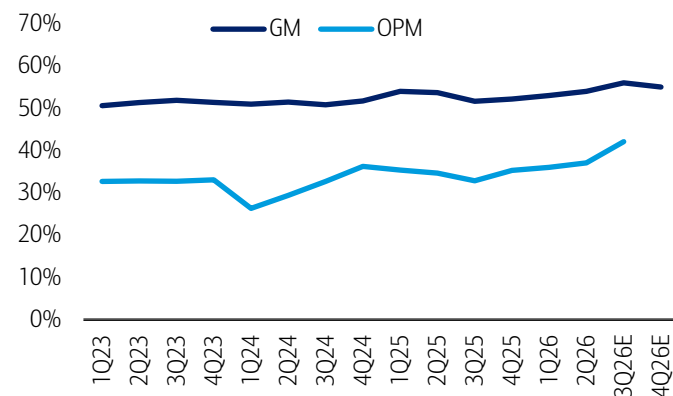


Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 17: Strong margins well-observed in 2Q26 (GM 54%, OPM 37%) and management's guidance indicates robust margins in 2H (GM 55%+, OPM 40%+)

ASML – Gross margin and OP margin trend

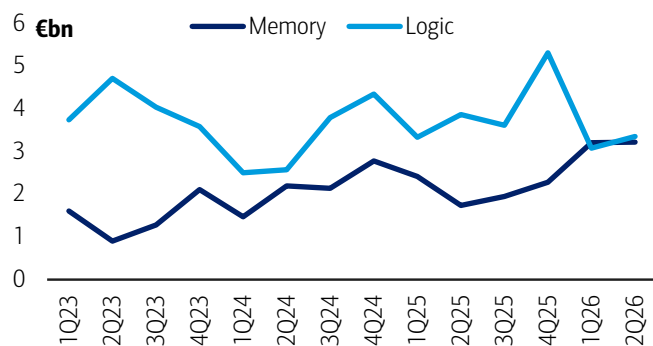


Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 18: Memory sales closely catching up to logic during the last three years on the back of DRAM super-cycle

ASML – Net system sales by application – memory vs logic (IDM+foundry)

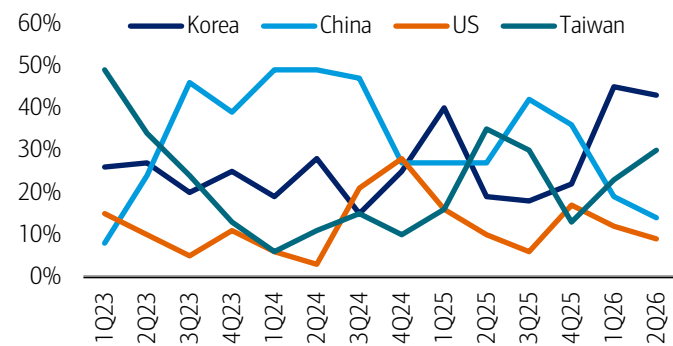


Source: Company

BofA GLOBAL RESEARCH

Exhibit 19: Korea accounted for 43% of ASML's net system sales in 2Q26, followed by Taiwan (30%), China (14%), and US (9%)

ASML – Net system sales by region



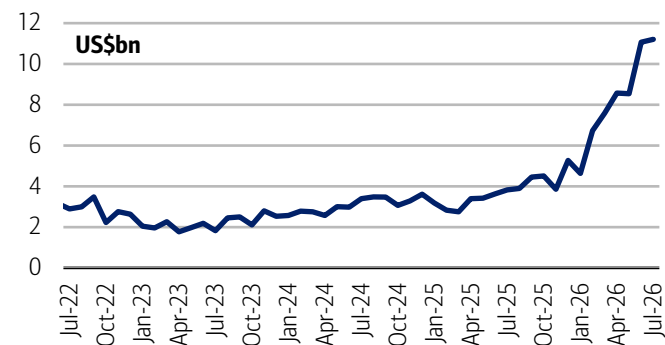
Source: Company

BofA GLOBAL RESEARCH

Korea exports and Taiwan sales

Exhibit 20: MoM growth rate decelerated, but still record-high in July (US\$11.2bn; +1.3% MoM)

Korea semis exports – First 10 days of month US\$bn

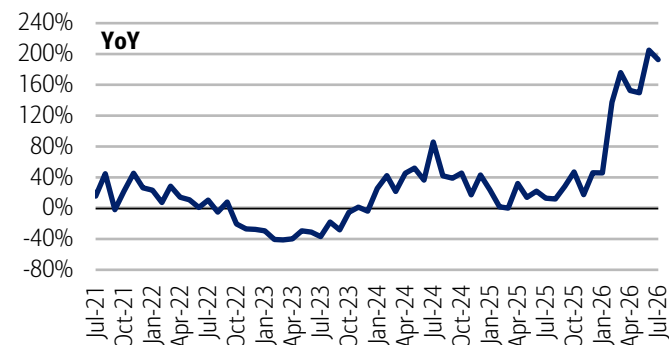


Source: MoTIR

BofA GLOBAL RESEARCH

Exhibit 21: YoY rebound still robust at +193% in July; already six consecutive months of triple-digit growth

Korea semis exports – YoY change in first 10 days of month

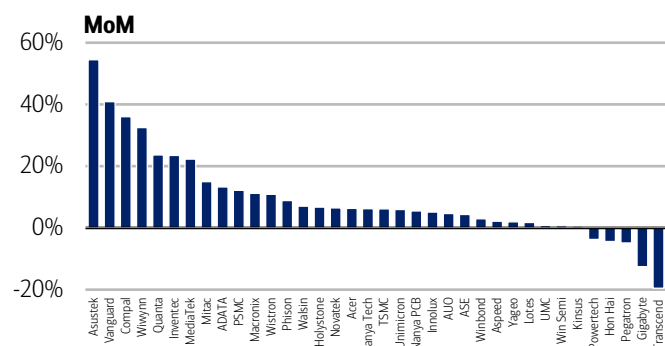


Source: MoTIR

BofA GLOBAL RESEARCH

Exhibit 22: Robust MoM growth led by logic/foundry/ODM names including Vanguard (+41%), MediaTek (+22%), Quanta (+24%)

Taiwan tech companies' MoM monthly sales (Jun-2026)

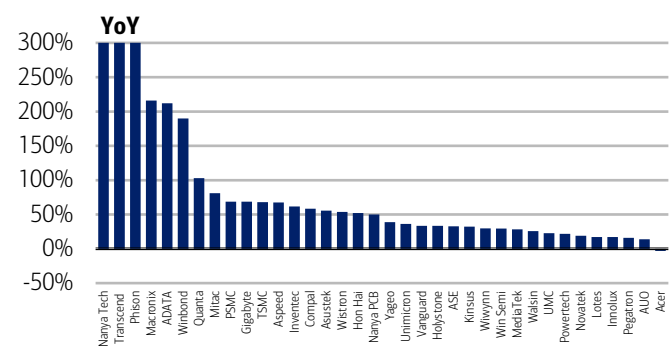


Source: Companies

BofA GLOBAL RESEARCH

Exhibit 23: Exceptionally strong YoY rebound led by memory names such as Nanya, Phison, Transcend; TSMC also solid at +68%

Taiwan tech companies' YoY monthly sales (Jun-2026)



*Nanya Tech up 621%, Transcend up 382%, Phison up 301%

Source: Companies

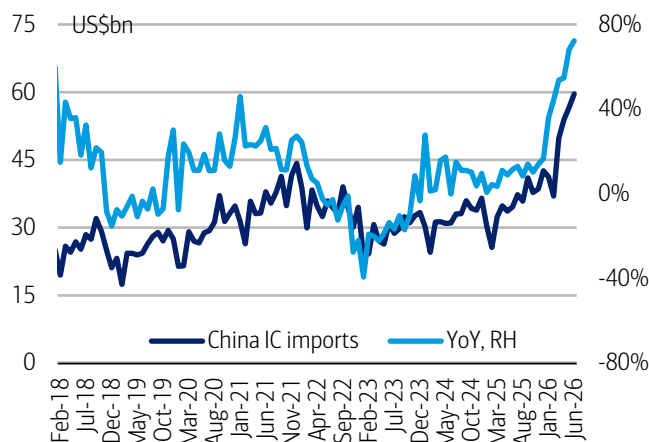
BofA GLOBAL RESEARCH



China IC imports and smartphone shipments

Exhibit 24: IC imports hit record high at US\$59.6bn in Jun, +72% YoY

China monthly integrated circuit (IC) imports and YoY

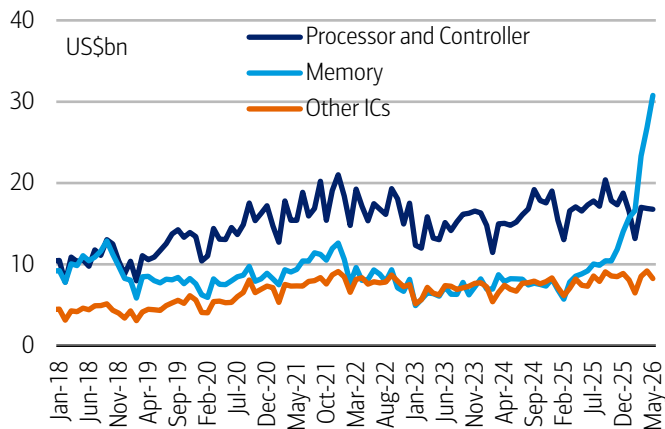


Source: China General Customs

BofA GLOBAL RESEARCH

Exhibit 26: Memory imports reached record high at US\$30.8bn in May, accounting for 54% of total imports

China IC imports by chip type

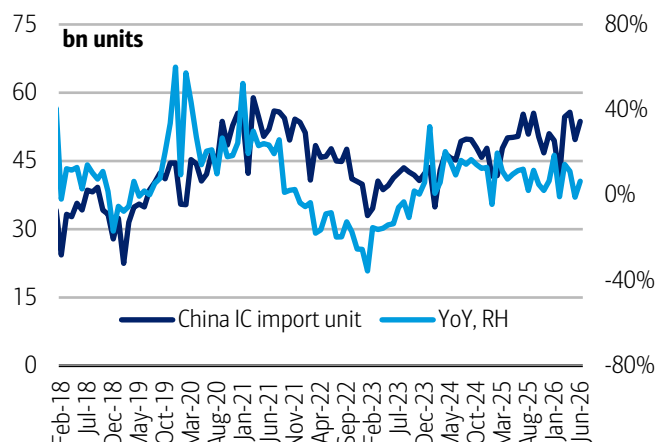


Source: China General Customs

BofA GLOBAL RESEARCH

Exhibit 25: China IC imports reached 53.7bn units in Jun, +7% YoY

China monthly integrated circuit (IC) imports volume and YoY

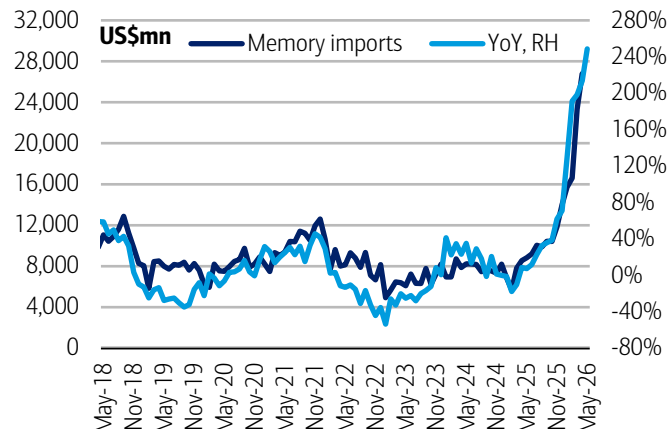


Source: China General Customs

BofA GLOBAL RESEARCH

Exhibit 27: Memory monthly imports grew 249% YoY growth during May

Monthly memory chip imports to China and YoY

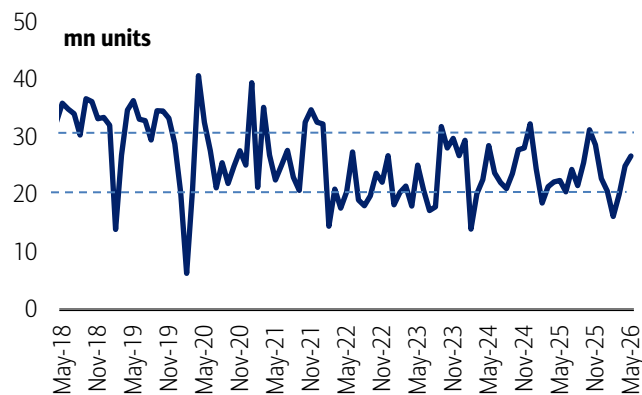


Source: China General Customs

BofA GLOBAL RESEARCH

Exhibit 28: China smartphone shipments were 26.8mn units in May; well recovered but sustainable growth not yet confirmed

China smartphone – monthly shipment

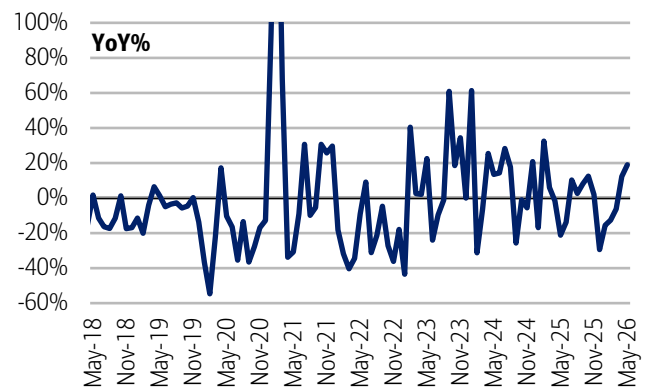


Source: MIIT/CAICT, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 29: May smartphone shipment +19% YoY despite DRAM shortage

China smartphone – monthly shipment YoY

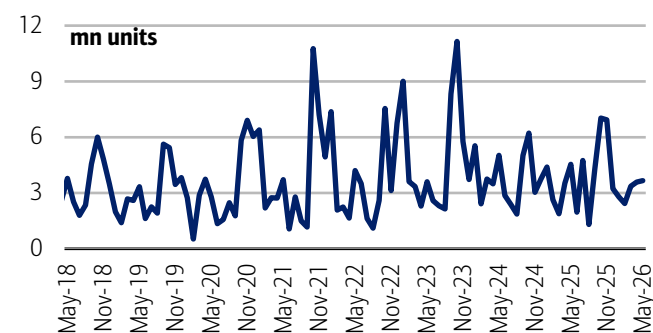


Source: MIIT/CAICT, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 30: Implied smartphone shipment by foreign brands 3.7mn units in May

China smartphone – foreign brands' shipment

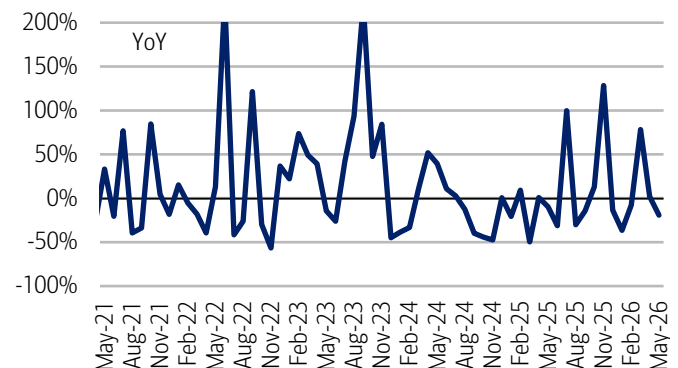


Source: MIIT/CAICT, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 31: Foreign brands shipment dropped 19% YoY in May

China smartphone – foreign brands' shipment YoY

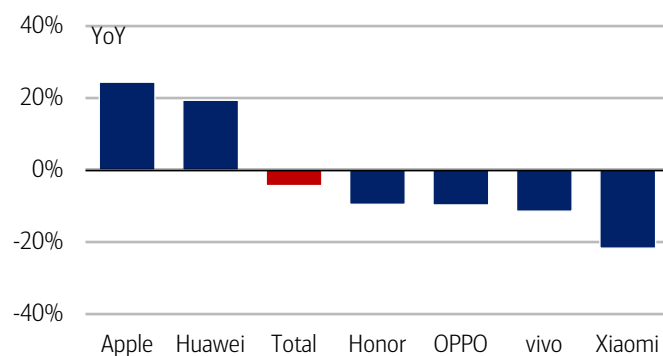


Source: MIIT/CAICT, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 32: China's smartphone shipments down 4.3% in 2Q26, with 19-24% YoY growth for Apple and Huawei while 10-22% decline for other major brands

China smartphone shipments YoY growth

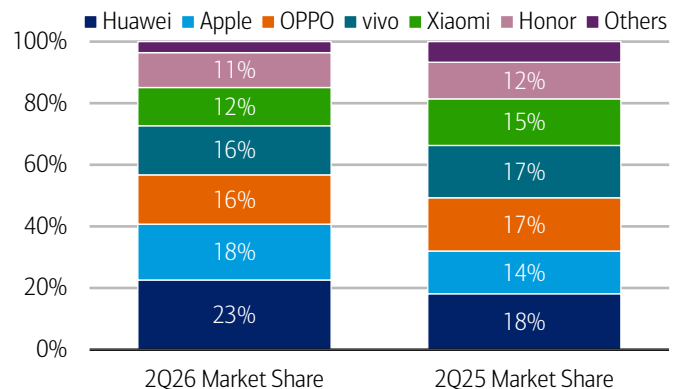


Source: IDC

BofA GLOBAL RESEARCH

Exhibit 33: Huawei saw clear market share gain (23% in 2Q26 vs 18% in 2Q25, followed by Apple

China smartphone market share



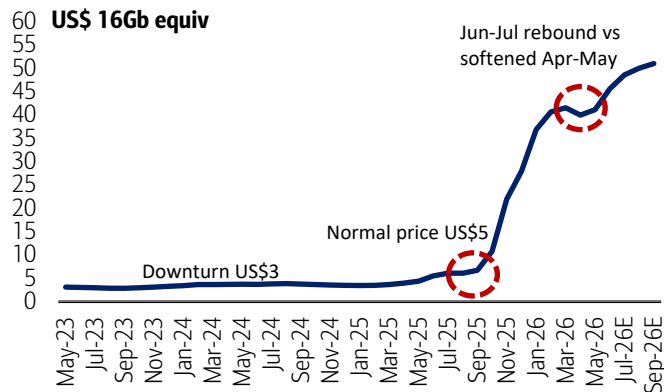
Source: IDC

BofA GLOBAL RESEARCH



Memory price forecast and spot price short term trend

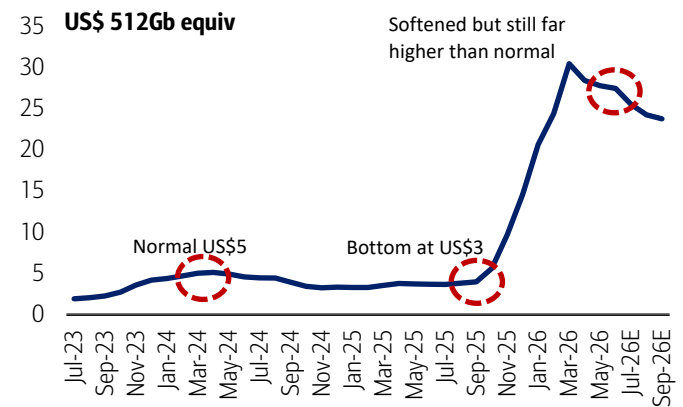
Exhibit 34: Well rebounded in June-July following April-May corrections; summary rally also possible given tight spot supply
 DRAM spot price outlook - US\$/16Gb equiv



Source: DRAMeXchange, BofA Global Research estimates

BofA GLOBAL RESEARCH

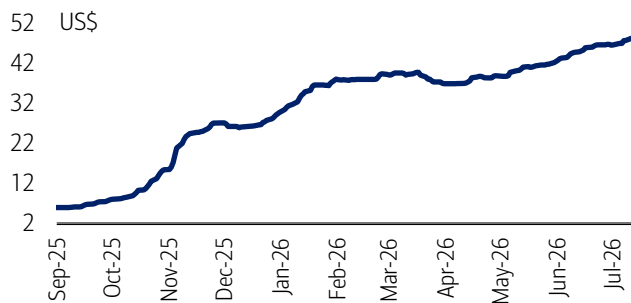
Exhibit 35: Unlike DRAM, NAND spot price has been corrected through June-July; further correction possible through summer but no hardlanding and absolute price still far higher than normal
 NAND spot price outlook - US\$/512Gb equiv



Source: DRAMeXchange, BofA Global Research estimates

BofA GLOBAL RESEARCH

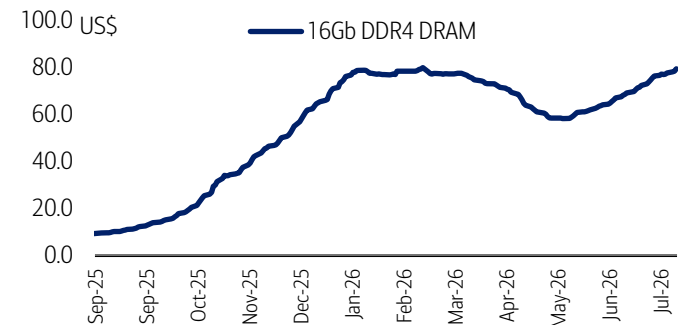
Exhibit 36: DDR5 prices reached new highs (\$49) in 1H-July, supported by strong AI-related demand and constrained supply as DRAM makers prioritize HBM production
 16Gb DDR5 spot price – daily, Jul'25 - Jul '26



Source: DRAMeXchange

BofA GLOBAL RESEARCH

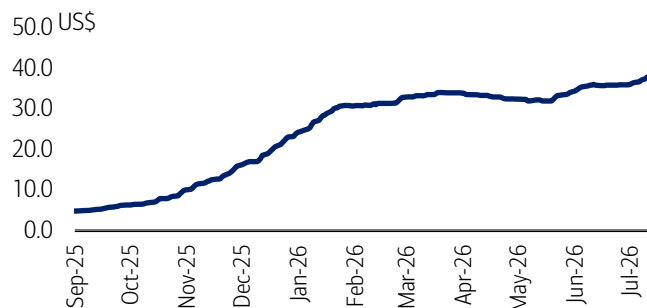
Exhibit 37: DDR4 prices remained near cycle highs (\$80), supported by supply cuts and customer restocking, with a brief correction in May before recovering through June and 1H-July
 16Gb DDR4 spot price trend, Jul'25 - Jul '26



Source: DRAMeXchange

BofA GLOBAL RESEARCH

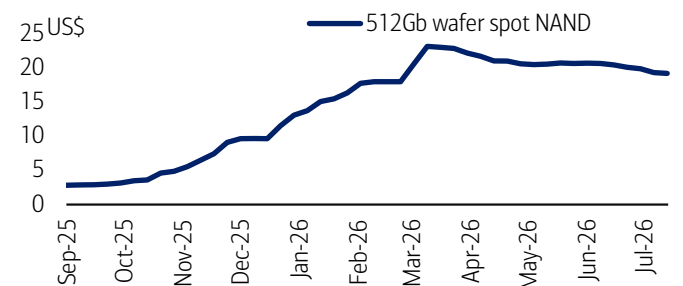
Exhibit 38: 8Gb DDR4 prices remained firm near highs, driven by production exits and inventory replenishment
 8Gb DDR4 spot price trend, Jul'25 - Jul '26



Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 39: NAND wafer pricing rose to elevated levels, through 4Q25-1Q26, on tight supply and stronger SSD demand, but has flattened/slightly softened through 2Q26 and 1H-Jul
 512Gb NAND wafer spot price – weekly, Jun'25 - Jun '26



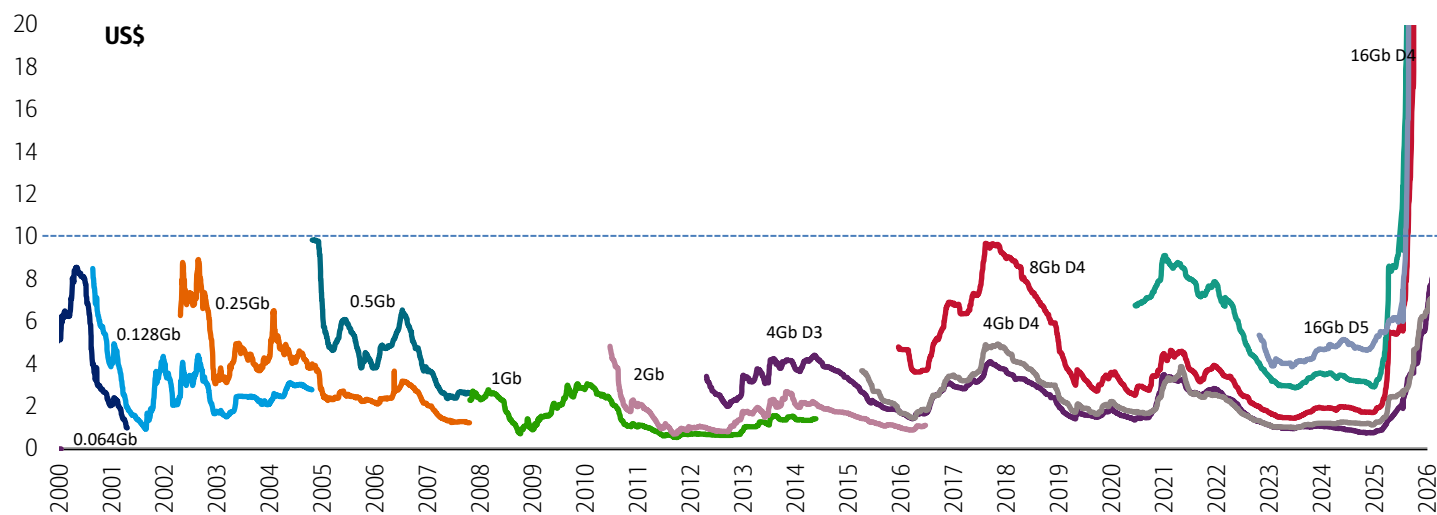
Source: DRAMeXchange

BofA GLOBAL RESEARCH

Memory spot/contract price long term trend

Exhibit 40: DRAM spot prices rebounded through June after softening through April–May, following a strong rally from September 2025 to January 2026. Prices have reached multi-decade highs, with 16Gb DDR5 at ~\$49 and DDR4 at ~\$80, driven by supply reallocation toward HBM and server DRAM—well above the prior ~\$10 peak in October 2017.

DRAM spot price – long-term trend (2000-2025)

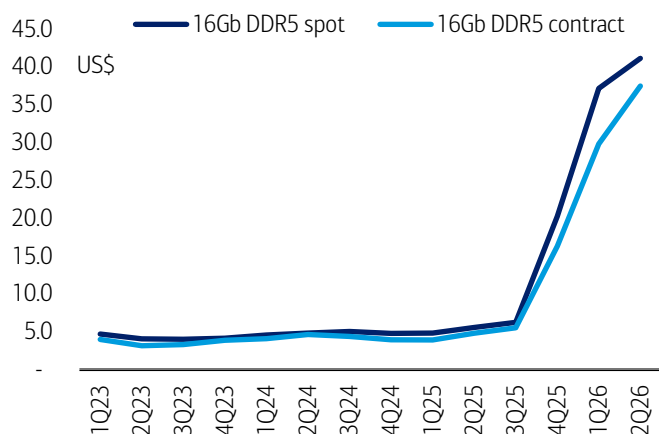


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 41: DRAM spot and contract prices climbed to record highs of ~\$35–40, with moderate growth in 2Q26 following a strong rally in 4Q25 and 1Q26.

DRAM spot and contract price - quarterly average trend

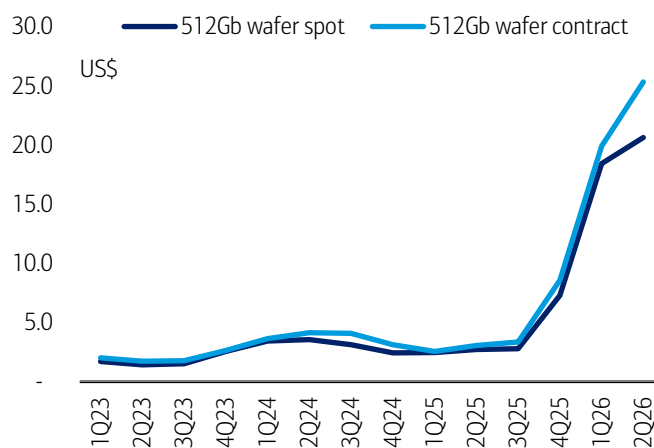


Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

Exhibit 42: NAND wafer contract prices have surged significantly ahead of spot prices, with both now at record-high levels

NAND wafer spot and contract price - quarterly average trend



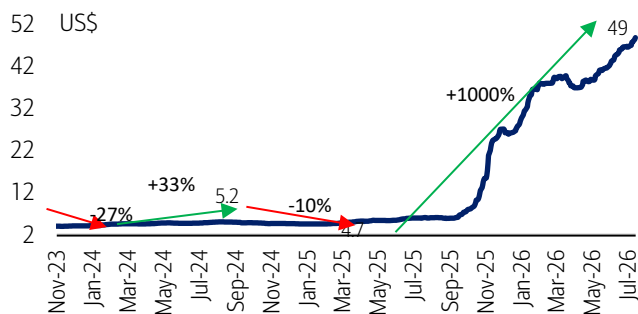
Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH



Exhibit 43: Prices rebounded through May/June/1H-Jul, reaching an all-time high (\$49). Uptrend was driven by strong gains in Oct (+70%), Nov (+60%), Jan (+25%), before moderating into a more subdued trend during Feb–Apr

16Gb DDR5 spot price – daily, Nov '23 - Jul '26

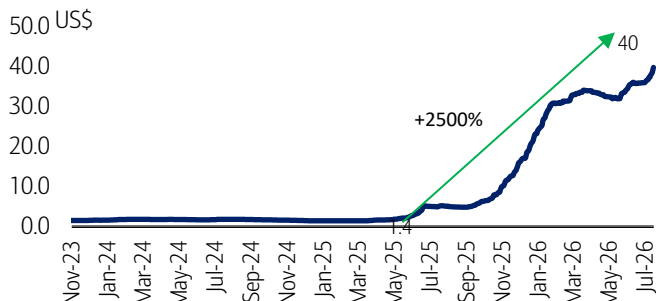


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 45: 8Gb DDR4 spot prices reached a record high of ~\$40 in 1H-July, well above the prior ~\$10 peak in October 2017, driven by ongoing supply cuts from major vendors

8Gb DDR4 spot price trend, Nov '23 - Jul '26

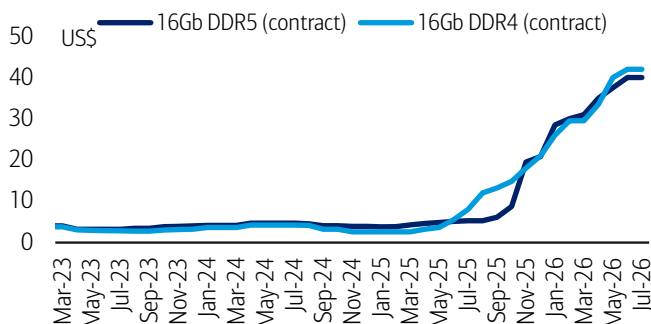


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 47: DDR4 and DDR5 (16Gb) contract prices are similar at US\$35-\$40 levels – DDR5 price premium no longer exists due to DDR4 shortage

16Gb DDR5 vs 16Gb DDR4 contract price trend, Mar '23-Jul '26



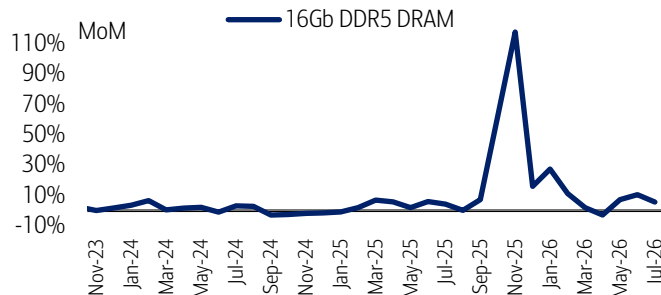
Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 44: Modest uptick observed in May/Jun/1H-Jul (+6-10%), following a sequential slowdown from Feb (+10%) to flat momentum in Mar and a minor decline in Apr, after the strong Jan (+25%) surge and the sharp Oct–Nov-25 rally (+60–120%)

16Gb DDR5 spot month-average price – MoM change, Nov '23 - Jul '26

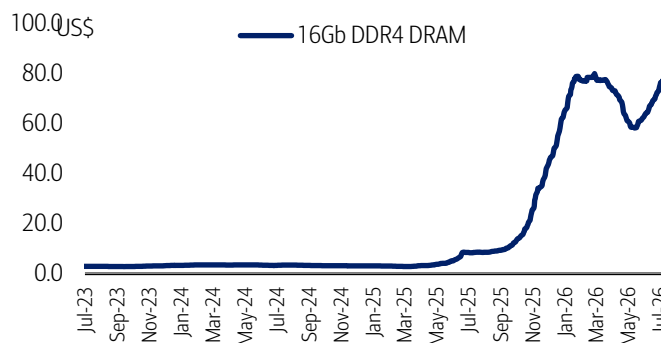


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 46: Prices rebounded to record highs in late May–1H-July after a ~25% April–mid-May correction, remaining ~2000% above the October 2025 low (~\$3)

16Gb DDR4 spot price trend, Jul '23 - Jul '26

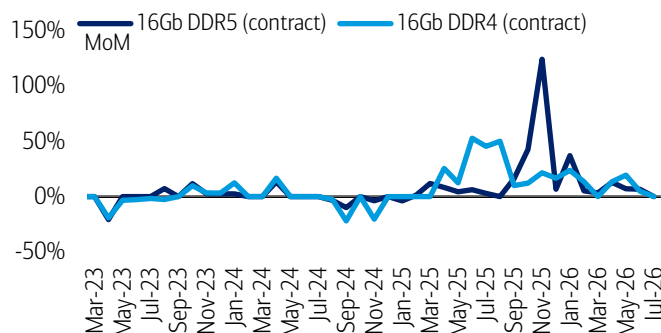


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 48: DDR5 and DDR4 contract prices expected to be flat MoM in Jun/Jul vs up +10-15% MoM each in Apr/May-26; price strong through 2H25 and early 2026

16Gb DDR5 vs 16Gb DDR4 contract price change - MoM, Mar '23-Jul '26



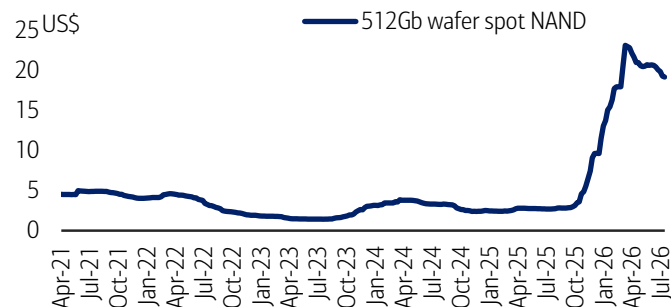
Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 49: NAND spot prices remained broadly stable to slightly weaker through 1Q26 and 1H-Jul, but still stand more than 50% YTD and nearly 8x higher than the Feb-25 trough

512Gb NAND wafer spot price – weekly, Apr '21 - Jul '26

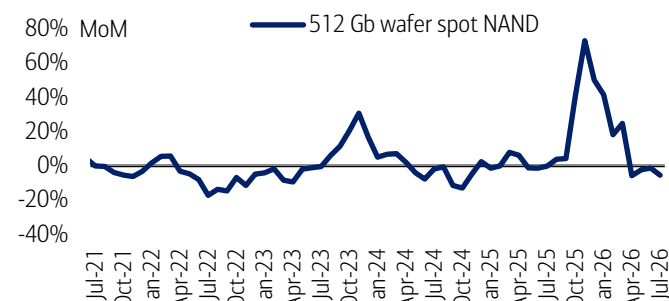


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 50: Prices were largely flat in June and slightly lower in 1H-July, following the April-May correction, compared with gains of 15–20% in Feb-Mar 2026 and 40–70% MoM increases during Oct 2025-Jan 2026

512Gb NAND wafer spot average – MoM change, Jul'21 - Jul'26

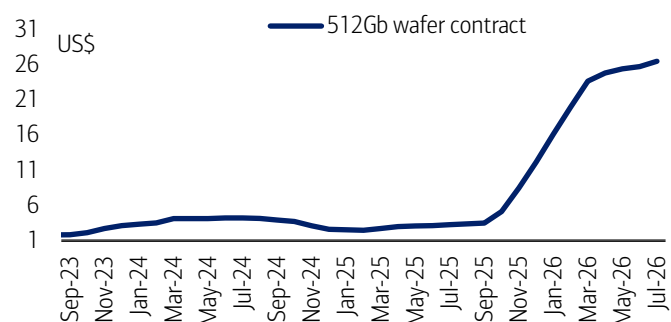


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 51: Current contract price at ~\$26, around 10x vs Feb-25 bottom of \$2.5; prices significantly up in 4Q25/1Q26 and remained broadly stable thereafter

NAND wafer contract price trend, Sep '23-Jul '26



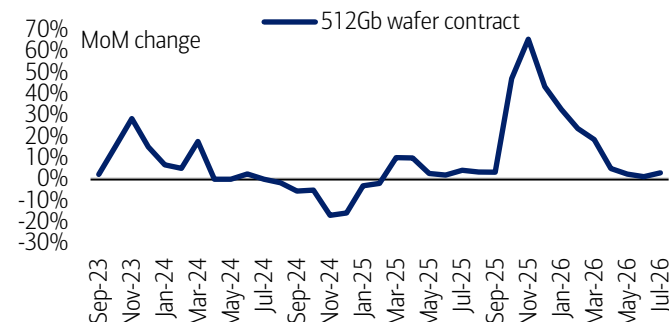
Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

Source: TrendForce, DRAMeXchange, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 52: Apr/May/Jun/Jul price only up +1-5% MoM each following Jan/Feb/Mar-26 rally (up 20-30% MoM) and Oct/Nov/Dec upturn (up 40-60% MoM)

MoM change of NAND contract price (512Gb wafer Sep '23-Jul '26)



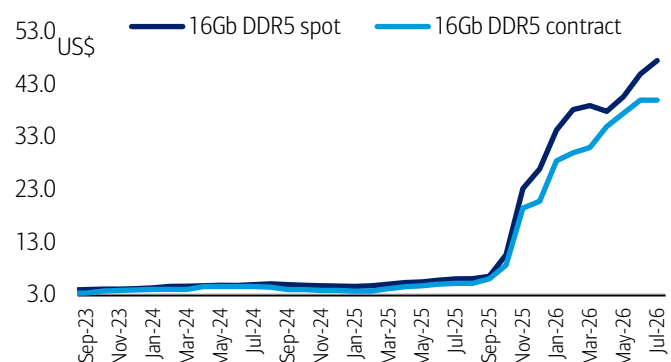
Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

Source: DRAMeXchange, TrendForce, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 53: 16Gb DDR5 spot and contract prices are in the range of US\$35-40 vs historical range of US\$3-5

16Gb DDR5 spot vs contract price trend (US\$), Sep '23-Jul '26



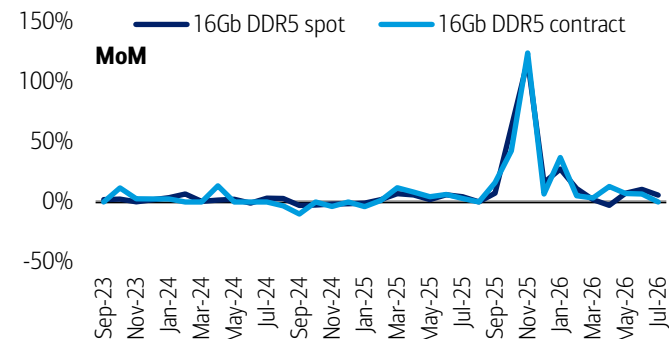
Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

Exhibit 54: Jun-26 spot and contract prices turned positive again after Feb-Mar moderation, while Jul spot price rose, contract remained stable; prices had witnessed a sharp Oct-Dec'25 rally

16Gb DDR5 spot vs contract price trend – MoM change, Sep '23-Jul '26



Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

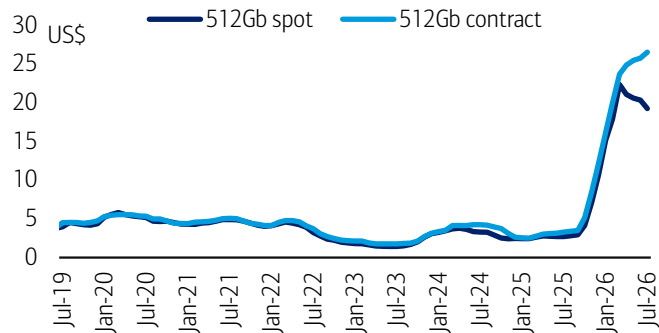
Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH



Exhibit 55: Current NAND spot and contract prices are a few times higher than 2025 summer level

512Gb NAND wafer spot vs contract price trend (US\$), Jul '19 - Jul '26

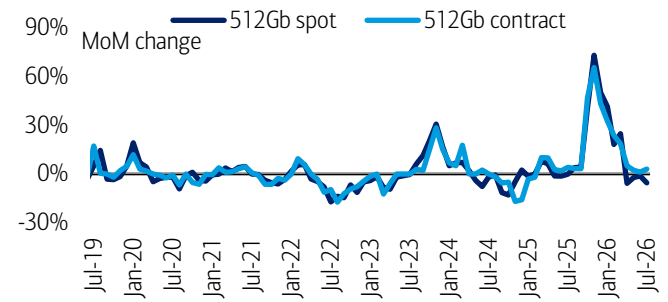

 Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

Exhibit 56: NAND spot and contract prices normalized in Apr/May/Jul/Jul after strong rally during Oct-25 – Dec-25 (up 40-50% per month) and Jan-Mar-26 (20-30% per month)

512Gb NAND spot vs contract price trend – MoM change, Jul '19 - Jul '26

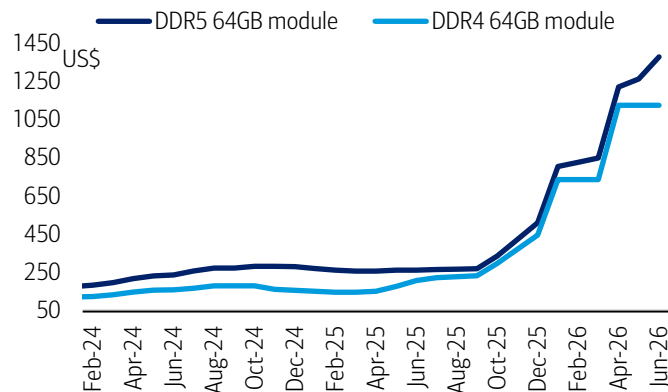

 Contract price update for Jul-26 is TrendForce estimate released on 15th Jul

Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

Exhibit 57: Current price of 64GB DDR4/DDR5 modules hit all-time high of more than \$1,000 level (DDR5: US\$1,400, DDR4: US\$1,100)

Server DRAM contract price trend – DDR5 vs DDR4 modules, Feb '24-Jun '26

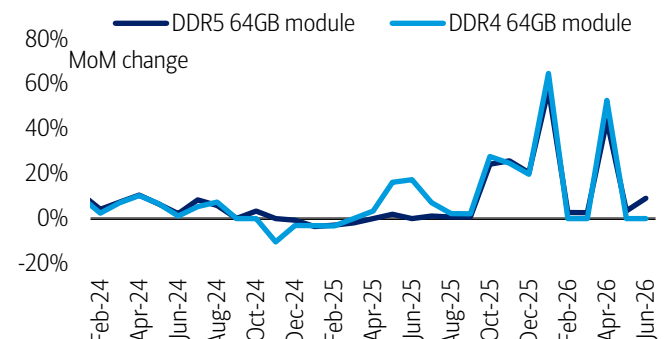


Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 58: DDR5 server contract prices recovered in June, while DDR4 remained flat in, following a sharp 40–50% MoM increase in Apr-26 and a similar 50–60% surge in January, after largely stable pricing conditions during Feb–Mar

MoM change of server DRAM contract prices, Feb '24-Jun '26

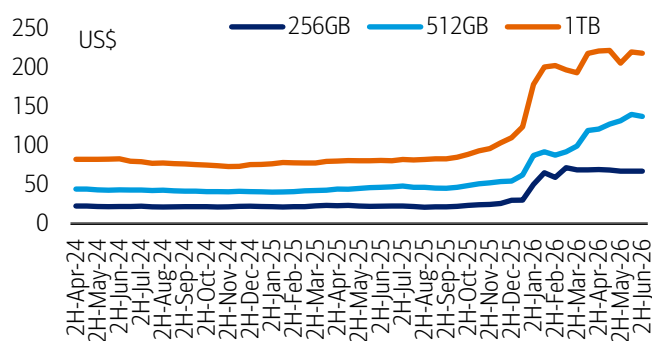


Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 59: 2H-Jun SSD price slightly down vs 1H-Jun rebound; prices witnessed a sharp surge in 1Q26 and April (DRAMeXchange), following a gradual upward trend throughout 2025

Client SSD price trend – mostly for PC (not server), Apr '24 - Jun '26



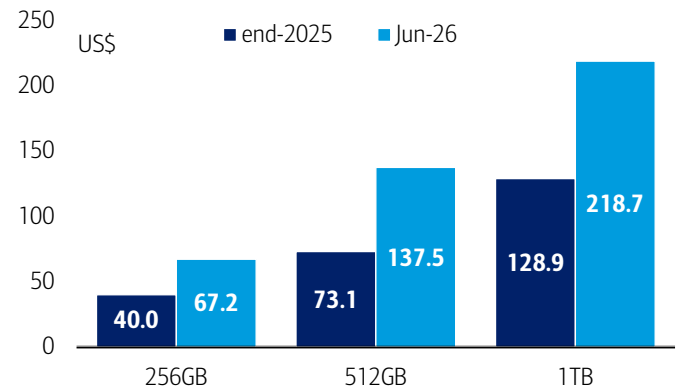
Note: SSD prices are as of 28 Jun 2026, reported by DRAMeXchange. TB = terabyte.

Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 60: Jun prices have doubled compared to end-2025 levels, while the 2025 increase was more modest at around 35–40%

Client SSD (for PC) price comparison – current vs. end-2025



Note: SSD prices are as of 28 Jun 2026, reported by DRAMeXchange

Source: DRAMeXchange

BofA GLOBAL RESEARCH

Valuation and stock performance

Exhibit 61: Memory stocks have been so volatile due to earnings miss concern vs solid fundamentals (high margin) and low P/E multiples

Valuation comparison among memory and semiconductor supply chain stocks

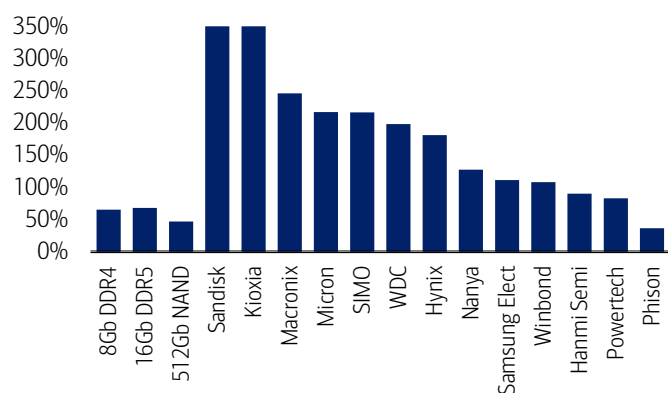
	Ticker	Rating	Price (Local)	Mcap (\$bn)	P/E			P/Book			EV/EBITDA		EV/Sales		ROE			Div. yield	
					FY25	FY26E	FY27E	FY25	FY26E	FY27E	FY26E	FY27E	FY26E	FY27E	FY25	FY26E	FY27E	FY26E	FY27E
Memory/DRAM																			
Samsung	SSNLF	B-1-7	273,500	1,001.3	41.7	6.3	4.8	4.4	2.6	1.8	3.6	2.8	2.2	1.8	10.8%	51.8%	43.5%	1.1%	1.8%
Micron	MU	C-1-7	904.28	1,021.3	109.1	12.4	6.4	18.8	7.5	3.9	9.9	5.2	7.6	4.3	18.8%	87.8%	81.0%	0.1%	0.1%
Nanya	NNYAF	B-1-7	439.00	47.1	207.2	7.0	5.2	8.9	3.3	2.2	4.4	3.1	3.4	2.3	3.9%	67.9%	52.0%	4.6%	5.7%
Memory/NAND																			
Kioxia	XIYUF	C-1-9	62,110	209.6	60.7	6.7	5.2	24.1	5.3	2.6	4.1	3.2	3.4	2.6	51.9%	128.2%	67.2%	0.0%	0.0%
Sandisk	SNDK	C-1-9	1,615.00	239.2	n/a	23.4	6.9	26.0	12.7	4.2	17.9	5.1	11.5	4.4	4.3%	76.0%	96.8%	0.0%	0.0%
Western Digital	WDC	C-1-7	513.84	188.1	n/a	51.6	27.4	35.4	15.2	10.1	34.9	19.0	14.6	10.0	22.5%	42.9%	46.5%	0.1%	0.1%
SIMO	SIMO	C-1-7	293.40	9.9	80.6	32.6	27.6	11.7	8.9	7.2	24.2	19.1	5.7	4.8	15.3%	31.9%	29.6%	0.8%	1.0%
Phison	PISNF	C-1-8	1,980.00	13.6	49.9	5.5	9.4	7.2	3.7	3.1	4.1	6.6	1.7	1.5	16.0%	89.2%	36.0%	9.6%	5.8%
US/Taiwan semis																			
NVIDIA	NVDA	C-1-7	212.50	5,142.5	46.7	23.4	16.0	33.6	17.5	11.0	19.5	13.6	13.2	9.3	94.3%	95.9%	80.3%	0.4%	0.5%
AMD	AMD	C-1-9	529.14	862.8	126.9	70.6	39.9	13.7	11.6	8.9	50.9	30.7	17.3	11.2	11.3%	18.1%	26.3%	0.0%	0.0%
Broadcom	AVGO	C-1-7	394.28	1,875.8	57.8	34.0	22.0	23.2	17.8	11.4	27.0	17.2	17.9	11.8	45.3%	61.3%	65.5%	0.7%	0.7%
Intel	INTC	C-1-9	102.99	517.6	239.5	97.2	59.9	4.1	3.8	3.5	25.4	19.3	8.5	7.4	1.7%	4.1%	6.2%	0.0%	0.0%
Arm	ARM	C-2-9	277.01	295.9	156.5	133.8	97.5	35.7	28.2	21.9	105.6	76.0	49.8	37.2	25.0%	23.6%	25.5%	0.0%	0.0%
Qualcomm	QCOM	B-3-7	177.98	187.6	14.8	16.7	16.2	8.8	7.1	6.7	13.8	13.9	4.8	4.6	56.0%	47.6%	41.7%	2.0%	2.0%
TSMC	TSMWF	B-1-7	2,470	1,990.0	37.3	23.1	16.1	11.8	8.3	5.7	15.3	10.8	11.4	8.1	35.4%	42.1%	42.0%	1.0%	1.0%
MediaTek	MDTKF	B-1-7	3,700	184.4	55.9	56.9	27.1	14.8	15.7	10.8	44.3	21.9	9.0	5.4	26.4%	26.6%	46.8%	1.4%	2.9%
Equipment																			
ASML	ASMLF	B-1-7	1,549.40	687.8	62.5	37.3	26.6	30.7	21.8	16.6	29.2	21.2	12.8	10.1	50.6%	68.0%	70.3%	0.7%	1.0%
AMAT	AMAT	B-1-7	579.43	460.0	61.5	47.7	35.7	22.5	15.7	11.0	40.7	30.5	13.9	11.1	38.6%	39.0%	36.0%	0.4%	0.4%
Lam	LRCX	C-1-7	335.43	419.5	81.2	59.4	44.4	42.5	36.7	26.4	48.8	36.5	18.2	14.2	57.9%	66.9%	68.9%	0.3%	0.4%
TEL	TOELF	C-1-7	70,890	204.6	56.7	45.4	30.0	15.7	13.2	10.4	31.0	21.2	10.1	7.7	29.3%	31.5%	38.7%	1.1%	1.7%
Hanmi Semi	HNSIF	C-1-7	265,000	15.6	117.5	78.3	43.8	36.6	26.9	17.9	59.6	32.6	30.0	17.8	34.8%	39.4%	48.9%	0.4%	0.8%
BESI	BESVF	B-1-7	244.00	22.6	144.7	56.6	33.9	46.3	33.9	24.8	43.0	27.0	19.4	13.8	29.2%	69.4%	84.6%	1.6%	2.8%
ASMPT	ASMVF	C-1-7	177.50	9.5	75.1	47.8	28.8	4.6	4.4	4.0	27.3	17.8	4.1	2.9	6.5%	9.4%	14.5%	1.1%	1.4%

Source: Company reports, BofA Global Research estimates; Mcap = market capitalization, ROE = return on equity, Div = dividend. RSTR = Restricted. Price as of latest close.

BofA GLOBAL RESEARCH

Exhibit 62: Memory stocks corrected sharply in 1H-Jul after a strong 1H26 rally; however, prices still remain elevated year-to-date, led by NAND and HDD with sustained strength in DRAM, underpinned by robust AI-driven demand

Memory companies – 2026 YTD stock performance comparison



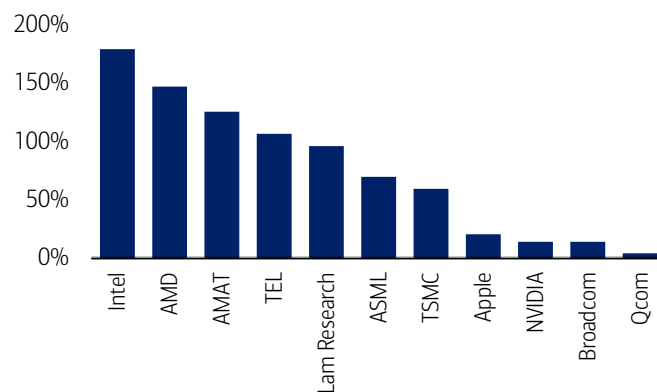
Sandisk/Kioxia shows above 500% YTD return

Source: Bloomberg, DRAmeXchange

BofA GLOBAL RESEARCH

Exhibit 63: CPU names such as Intel and AMD clearly outperforming other US big tech names such as NVIDIA/Apple/QCOM

Global major tech companies – 2026 YTD stock performance comparison



Source: Bloomberg

BofA GLOBAL RESEARCH



Disclosures

Important Disclosures

BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.

Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.

The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.

Officers of BofA or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.

Refer to [BofA Global Research policies relating to conflicts of interest](#).

"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.

Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:

BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofam.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSF); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority. This information has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSF; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the [Electronic Communications Disclaimers](#) for further information.

This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.

This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives,

financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are largely unregulated. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities that may have been recommended in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments discussed in this report.

BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.

In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQDatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS or any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.

Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.

Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.

The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.

All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.

Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.

In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.

Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.

